ActInf Livestream #030.2 ~ “How to count biological minds: symbiosis, the free energy principle..."

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hello everybody welcome to active lab live stream number 30.2 today it is october 12th 2021 and we're having our second discussion group discussion on the paper of matt sims how to count biological minds symbiosis the free energy principle and reciprocal multi-scale integration welcome to the active inference lab we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links here on this page this is a recorded and archived live stream so please provide us with feedback so we can improve on our work all backgrounds and perspectives are welcome here and we'll be following good video etiquette for live streams at this short link you can see some of the past present and future live streams in the lab and in the first tab you'll see the tuesday numbered live stream so every tuesday we have a participatory group discussion two weeks per paper and then there's the dot zero video which is just sort of a contextualizing video and then these other tabs like guest stream model stream and math stream are opportunities for just one-off events on any topic that people think is relevant for the lab so get in touch if you want to join one or co-organize one of these streams today in 30.2 we're going to be continuing to learn and discuss this awesome paper by matthew sims how to count biological minds we're just having fun and discussing last week we had matt join which was a great discussion we look forward to everyone's input and if you're watching in the live chat or if you're leaving a comment later look forward to continuing the discussion today we're pretty much just going to introduce ourselves and say hello and then we will go wherever we feel relevance and explore different questions people can ask about a figure or a section of the text and then also blue and others have prepared some discussion questions to get into so we can start with the warm-up and introductions so i'm daniel and i'm a researcher in california i think the thing i'm excited about today just like with many dot twos is taking a lot of these cool ideas that came up in the paper and in the dot zero and the dot one and then synthesizing and consolidating and then asking okay well now what next how do we go from this conversation theory users and resources reciprocal multi-scale integration and then apply it to different systems so i think that will be a fun topic and i'll pass to stephen thank you yes steven here i'm in toronto on a nice sunny day and um i'm really interested as well actually in these building on these ideas i'm really excited that the folk psychology and folk biology was brought up and i think that that gives permission i think to build that further in and maybe extends into how our phenomenology connects to our folk psychology because our folk psychology allows us all to be inherent psychologists of ourselves maybe so that's interesting and we just enjoy it analyzing our own reality so i'm going to pass it over to dave um yep i'm in the mountains of the philippines and uh it's the end of the rainy season and so we'll have no weather until next year um i am pumped up about the multiple disciplines that are coming together in the discussion of the paper that we're doing now and also about bobby azarian who is not afraid to jump down into the trenches and face the lunatics with their machetes and and even less pleasant uh arms um and getting a note together to him to suggest some topics that we might want to talk about well two two things one off a question about bobby but first was when you said there's no weather it's kind of like what stephen said about folk psychology no experience you know just like well now i'm awake and now experience is going to stop it's kind of like weather stopping like weather is always happening it's always changing in different ways and our experience is also always happening and always in different ways but just like weather sometimes only gets studied or discussed in the context of catastrophe or pathology or of extremely unusual events there's this like baseline weather that's just the climate and so that's very interesting about our day-to-day felt realities and then how everybody is in the driver's seat with respect to their experience so sort of the mundaneness of weather the mundaneness of experience but also it's something that can be really unpacked stephen yeah exactly the mundaneness is interesting especially when you think of the priors that set things up because um in some ways it's like the psychologist in psychology gets to decide on the modeling that's going to be used to look at people whereas in folk psychology this is the implicit modeling that we are framing ourselves around and of course the same with the folk biology which this paper brings up and maybe the folk physics and whatever so these kind of implicit priors um you can't get away from them and maybe in a way it brings us into this interest in are those priors how much of them are psychological or are they more like practice based phenomenological questions so it's an interesting balance there cool and then the second question so bobby um azarian is going to be doing a guest stream in november we have a few guest streams coming up so um there's multiple but dave what made you excited about his presentation or work with the universal bayesianism well it's not so much that particular approach it's that he's been doing a lot of work on cult formation uh cult dynamics the growth of violent cults especially in the us right now and that directly ties into philip guerin's work on the uh neuroscience of delusion okay very interesting and so i'm just showing on the screen those are the dates november 19th is um with bobby but we'll have um greg enriquez on the 11th or on october 22nd for number 11 then martin boots for 12 on october 25th adam saffron on november 8th and then november 19th so cool guest streams coming up and of course everyone's always welcome to join so let's return yeah i didn't notice you've gotten books great i am very pleased yeah sometimes people just sneak in there you know um okay so let's return to 30.2 and uh i'm sure blue will join so let's just save some tasty pieces for her that she added in and then otherwise i think we can maybe let's start with these questions on seven so blue in absentee asynchronous blue our favorite shade contributing to the discussion let's see what we think and if we can connect it to some of the ideas or sections or figures of the paper at any given time can agents act autonomously as long as there is a defined prior and then i think maybe that's connected to the second question is there a difference between a cognizer and an autonomous agent so how are the terms of cognition agent slash agency and autonomy how do these terms connect like if we imagine a three-dimensional space and we think okay you could have you know ten zero zero you have ten cognition zero agency zero autonomy what is the space of the possible of cognition agency and autonomy and then how are these different terms linked are one out of necessity uh leading the other or lagging the other so stephen and then dave i mean this question about a defined prior i mean the other option is is it if they don't need a defined prior and they can build up inference [Music] process without having that prior so i i'm not sure which way that might be the most important is it is the autonomous agent obviously we're assuming the agent can survive you know this is not a bolt of lightning or something so it's something that can have some sort of non-equilibrium steady state over time is it more about the autonomous agent being able to act and process its active inference independently of others and that's maybe more key maybe that's the sort of thing that happens with cancer as well you know um so i don't welcome blue we're just on slide seven working through some of your interesting questions dave yeah on that the first question um in the dialogue between christian and marx solms they use prior basically to mean an affect or the other brain stem resident biological imperatives and um i think some would tell you that except in anesthesia you know deep deep anesthesia and perhaps coma there's always operative priors because you really can't uh dissociate the um homeostatic goal seeking from the higher processes since of the higher processes are completely dependent on and shaped by the uh the homeostatic uh activities and i wouldn't even accept coma uh without qualification because of the i don't know if this research has been worn out but there are therapists who have simply proceeded under the assumption that deeply comatose individuals actually are taking something away from their experience and they're finding that in recovery from coma that yeah if you talk to people on a daily basis so you talk about interesting things or unresolved problems this shapes the way they're going to uh talk about their experience afterwards uh one thing that happens uh is that people people's thinking becomes extremely slow minutes rather than second and that if you actually speed up videos of the response of comatose individuals you'll see responsible just like if you slow down high resolution videos of infants you'll see that yes they're highly attuned to what's going on around them they're they do have their own priors that through which they're responding to events and actually shaping them in a healthy diet interesting so i think starting with uh with steven's point and then anyone can of course can just raise their hand or write a question live chat so the question about starting with the prior and whether we start with a very defined prior like a domain specific prior or whether that's learnt that is reminiscent of the difference between first order cybernetics and then higher order cybernetics like the first order cybernetics of chemistry is your priors on chemistry in the second order would be your approach to learning or your dynamics of flexibility around study habits or around updating your generative model of chemistry as well as the second order generative model of learning as we start to sort of meet in the middle with building active inference up describe and model these increasingly recursive and symbolic cognizers will start to see the movement from models that merely describe the mountain car into models of mental action higher order cybernetic models of mental action so that will be very cool and then um i think you mentioned the lightning bolt and so i was coming back to this this um trilemma or these three terms there's more terms could be added like um does lightning bolt have agency does it have autonomy well it sure seems to just act the way that it's apt to so it seems like an autonomous system it's not waiting for a bluetooth connection with something else so it's high autonomy and then does it have agency though and so how is there a difference and then cognition especially because the example of the lightning bolt it's electricity so if what we've been talking about with um professor levin and others is that like bioelectricity is sort of the fundamental that's where biotic and abiotic cognition come together or where they they blur so then is a lightning bolt cognitive so that was and then just a last comment was the speed of perception and cognition action well active inference is about integrated modeling of perception cognition and action so then we're in a quasi-computational framework there are computationalists among us people who think that cognition is computation as well as people who take more nuanced steps or reject that notion but this is kind of like our clock speed this is like the speed of that loop what's the tick speed of the generative model and that relates to the thickness of our temporal experience like flickering images at 30 per second are perceived as continuous so we don't know exactly what the clock speed is or whether a clock speed is even the apt metaphor but surely there are stimuli that are so close spatially or temporally we cannot tell the difference so it was just cool to hear like that different ages and different conditions might have different um parameters for their internal generative model and perception action loops blue and then anyone else so going back to whether or not a lightning bolt um has an agency it certainly seems that like if there is a defined fire like for a lightning bolt there's a defined like prior state that you know like the dielectric constant has to break down for the lightning bolt to be able to occur so as long as there's that defined prior then yes like the uh the lightning bolt acts autonomously but but does it have agency is like well what's the goal of the lightning bolt is it to like minimize the electricity build up in the cloud or like what would the goal of a lightning bolt be um and so that's kind of what you think about like even or what i think about when it comes to like cells okay the goal of a single cell organism is to maintain homeostasis at some you know set point or within some range so but a lightning bolt really what's the goal of a lightning bolt um and so i think about that i don't know so it does a lightning bolt have a goal and and that's kind of where the sense of agency comes into play for me um and like yeah if you define the goal like in a you know autonomous car like your goal is to safely transport people from a to b without running anyone over or getting smashed like or that's very simple i know but you know obeying all the traffic tools and so forth so you can establish a goal for an autonomous car but does the car come up with its own goal or is that like a human-driven goal and is the car then a tool for the humans so so there's like some fuzzy boundaries here that kind of always point toward um the life-mind continuity thesis which i think we probably haven't gone to yet but maybe we'll get there um where life has to happen at some degree for for cognition to occur maybe or maybe not very cool so the goal of the lightning bolt so let's think of a few you know physical systems we know and love like a ball rolling down to the bottom of a bowl it's minimizing its kinetic energy differential that's one description not to say that's the explanation but it's minimizing something by rolling to the bottom of the bowl without fail similarly when there's an electric charge buildup between the cloud and the ground or between two electric components inside of a electronic device there's like a zap of electricity that can also be understood in terms of minimizing some energy differential that's what's interesting about active inference is we're also thinking about minimizing differentials of information and so what stories are explanations or descriptions of the ball rolling down kinetically or the lightning strike electromagnetically and then are we going to have the exact same template of explanation for action of biological entities so that's a and where it is teleology and directedness like is the ball rolling into the bottom of the bowl is that end directed or is it just directed only by approximate teleology like a next step only so people say evolution has no end it's just a local optimization process for example so it's behavior along those lines um and then the second point then stephen and dave was like is the system of interest the primary cognizer or is it being used in an extended or an instrumental fashion wasn't sure on the answer there but that speaks to matt's paper with the users and the resources and so there's cases where you have like an adaptive active inference agent utilizing a mere active inferencer as a tool like you could be using a thermometer as a tool as you investigate the cave but then there's this other reciprocal case where actually the the primary cognizers are sort of adaptive on both sides so there's a lot to explore there stephen yeah the idea of things being either frozen at a certain state or having the ability to be a whole to stay in that state um i think about um well i mean on a big scale the sun right the sun has got an energy differential but um it's it's it's self-maintaining right because as it keeps going it more more material can be fused in the middle and it keeps so you get into the broader question of complex adaptive systems of which non-equilibrium steady-state systems are a subset so you've got this um you know there's an interesting point there and when you have something which dissipates and can't form a markovian sort of boundary um you so it will lose that but there are times when you can maintain that for instance um if you have a globular molten silica plunge it into cold water you can freeze the liquid state it's still a liquid but it's glass now so this same so it's maintaining that over longer periods of time but is it is it still not equal is has it has it formed is it just waiting to allow its energy gradient to dissipate or is it now managed to hold its state more concretely very interesting has a true pattern or structural integrity been reached or are we just sort of pausing or freeze freeze thaw on the energy differential just sort of make the ball made of velcro so it's still stuck up there dave yeah there's a hilarious tour de force by uh stuart bartlett life as a solution to the planet earth's dilemma of having too much heat and needing to get rid of it faster and he breaks down all these amazing processes culminating with the reverse krebs citric acid cycle as increasing the efficiency with which heat can be radiated by creating just the right kinds of molecules in enough concentration to catalyze that problem of boy it's hot in here this um it's kind of like the science fiction classification i don't know what class you know class one class two class three civilizations whether their energy capacity is like less than a star equal to a star multiple stars some thermodynamic perspectives on the emergence and the elaboration of life when we see living systems like schrodinger or scott david as entropic systems that are resisting dissipation and you realize that for every unit of order that is constituted inside of the markov blanket that more disorder must occur outside you know that it can't be um free lunch and it isn't going to be one to one either so you're making an excess of disorder which is what pulls chemical reactions forward when you order like for every you know one pound of muscle organized on your body more than one pound of muscle equivalence have to be dissipated and then it is actually this imperative that's thermodynamic for life to continue inventing potentially even increasingly seemingly inefficient mechanisms that's sometimes what i think about when someone talks about the energy use of cryptocurrency i think you know if you went into the cell and you said well we already have the enzyme making the forward reaction why do we need the one making the backwards reaction that's wasting energy it gives flexibility to the cell and it pushes the reactions forward and it gives capacity to regulate very adaptively so it's not always like we can just look at the um heat producing light bulb and say that that's the inefficient one so blue and then stephen i wonder too thermodynamically um is like life ultimately causing like the heat death of the universe right like are we using up all of the the thermodynamics free energy that's available um with the process of life like we are like in the scott david sense um you know entropy secreting right we're the only entropy i mean life is the only entropy minimizing thing that i can think of but i mean it's just ultimately going to lead to the lack of free energy thermodynamic free energy that's available to do work um which is just an interesting thing to think about and then also i was you know listening to some older live streams and maybe uh maybe dave actually probably has the knowledge to kind of clarify for me but i always um thought that informational free energy was fundamentally connected to thermodynamic free energy um is like in what someone had suggested um that was not the intention of info like of the free energy principle was never supposed to involve thermodynamics like initially um and i just wonder um is that like some abstract bridge that was formed the connection between infodynamic and thermodynamic um entropy or is that a real concrete connection and so for for me it's something i've always thought about as as like you know links through physics and statistical physics like those two things go together but is there a disconnect there that i'm perhaps not aware of and um was the free energy principle not meant to include thermodynamics so i would love to hear what um you guys think about that yeah dave go for it and then stephen yeah you know i think both entropy and the free energy principle are higher order metaphors or analogies at least um i'm guessing an entropy is roughly a third level analogy you have to go up to that level of ramification to cover the various cases but there are people working on that i think i sent you a paper by uh sean carroll on the thermodynamic uh interpretation of of what was it the free energy principle i just sent that a couple of days ago i don't know if you saw that i dropped it one of the chats dropped it in the corner of a labyrinth yeah i'm going to have to go back to buying borges i think he's i've left too much in his various lab room stephen yeah there's a couple of good points there um i suppose one thing in terms of life it's it's trying to stay in that golgi locked zone so there's times when it's trying to well in terms of heat anyway get rid you know reduce the heat and sometimes it might want to gain heat but it doesn't really care per se it's like the the niche itself will be have a greater fitness and if there's enough fluctuations over time and enough feedback loops in theory you'll sort of gain the carbon dioxide levels as best can serve the overall production of carbon mass in theory but in terms of the piece around it's really it's a good conversation around um active inference versus the free energy principle because the free energy principle or variational free energy jeffrey hinton's work and the recognition models and work that just takes you know perceptual data coming in with deep mind can work with reducing free energy information but when you bring action into the game so that's the interesting thing because then it's the action that brings in the energy in a way because by the action being more efficient the energy sort of becomes part of the landscape um whereas the perception piece you're kind of tied more to if there was no actual active influence and just bayesian influence it's kind of all information up and down but once you start bringing action into the game that's when it starts to get a bit more interesting it's a lot to think about information is usually framed more on the inference side than the action site we know there are information theoretic approaches to control theory and there's all kinds of connections that you know 2021 we can look back at and say that have been made but information is very aligned with knowledge and whereas thermodynamics brings in physical constraints energy and action like what is temperature if not a summary descriptor of the actions of molecules whereas information tends to be a little bit more abstracted and disembodied and thermodynamics even if it hasn't been connected to 4e cognition extended embedded and cultured etc it at least is closer to it like the the pressure volume relationships of a gas they relate to its context and to how it is actually enacting physics so then if we think about in informational physics first off how far can we get with bringing thermodynamics into information big topic and then where is control and action in that kind of a model stephen yeah this is interesting to bring in and also if you go back to gibbs free energy the way it was traditionally used i mean you basically have reactants and you have products which is basically where we were in terms of an equilibrium state and where we ended up and then there's a force that's going to drive that there's an enthalpic force i.e an energy change which will drive things and will succeed depending on the activation energy of whatever's in between which can be calculated and then there's this kind of entropic force which is basically cough cough um it's just inferred basically by what the reaction actually did but because you're only going from equilibrium state to equilibrium state whatever happens in the middle was just i don't know let's be that's basically if you if the truth be told it's just entropy right so but that's not it's not just this black box in this case it's actually all the stuff that's going on on the journeys is where active influence is sort of in the process so in tropy or thermodynamic entropy it's kind of interesting because there's there is somehow this enthalpic term in terms of energy that there's an entropic force they call it in chemistry which is like how the entropic shift sort of pushes the reaction and gives it a force which then combines with the enthalpic force which is basically just whether it's going to give off loads of heat or absorb heat oh sorry i should say energy absorbable you know or emit energy so this is where the the the i think it's not entirely clear quite how that might mix up in and maybe sometimes it's safe is to not make the analogy because it's non-equilibrium and non-equilibrium chemistry has only really started to happen in any sort of coherent way since the 1990s in terms of research it's all been equilibrium state equilibrium state and hand-waving in the middle thanks steven blue i um like whatever happens in the middle uh i only had abbreviated p chem and like this kind of makes me wish that i had more and you know then like a full semester of thermodynamics to to really like be able to screw these pieces together but but maybe they don't all actually screw together with a perfect fit so i mean i think about okay is it an exothermic or an endothermic reaction does energy go in or does energy come out and essentially all life is all of life is endothermic like some energy has to go in and where is that energy in life like because we exist in this non-equilibrium point like life isn't spontaneous and so where is this like storage happening and like in some like maybe it's a very abstract link but but i think about the energy that going that goes into creating life as information because it's essentially this dna blueprint that we all have that carries on from generation to generation that is the catalyst for the reactions that occur that create um so i that's just always i don't know how i think about it here's here's one image that you'll see on your shared screen so this relates to gibbs free energy i mean you mentioned it we've been talking a little bit about chemistry this is already pre-adapted to instead of be the gibbs free energy it's like the pain axis so we can quibble with the specific metaphor but the the piece that i think is interesting is the difference between the thermodynamically favored and the kinetically favored reactions so the green line is kinetically favored if there's a ball rolling it's easier to climb the b hill so it's easier to get on that there's less friction less of a activation energy cost and therefore it's kinetically more accessible to take the green route to get you to product d and then that a minus d is the delta free energy whereas the thermodynamically favored product which is like a to e this is like a complete combustion versus a partial combustion more energy is released there's a bigger delta g that product may not be as kinetically accessible and so this is like classic ochem chorus questions some reactions you want to do them really hot because you just want to get over a kinetic barrier and wherever you end up is going to be okay other times you want to carry out the reaction very slowly so that you can be sure that the only high energy balls that come across are only going to take this green road and so it doesn't like combust so you keep the reaction cold and then in this example it's already been connected to sort of behavioral economics like making a bad decision versus making a good decision you know you're choosing what to eat for dinner is it going to be something with a low barrier high kinetic accessibility but maybe not healthy but where do we transmute from the kinetic versus thermodynamics of chemicals to kinetics and thermodynamics of information and decision-making and action the closing the triangle would be like is there a energetic or a metabolic account for decision making like it literally uses more glucose in your brain to make taxing decisions and that would sort of close the loop in an interesting way so stephen then dave yeah this is so this is also quite useful diagram because i mean you'd have that kind of activation transition state as being the kind of the hill right so the higher the hill the more uh energy is needed to make that transition state become available but it is a case of it going from left to right within reason i mean there is that idea that things can move both ways but what's interesting is the more dilly-dallian to use this i know that's an english phrase um which i do a lot of but the more dilly-dallian is not necessarily the better thing there right because the more dilly dally and the molecules are doing the less they're going to get over the hump because it's really a case of this energy dynamic and entropy being maybe going to help um because because it in certain cases but normally it's a hindrance so something to slow it down but the difference is when you're dealing with variations see when you've got variational free now you can get information the dilly dalliand gives you variation in entropy and that can give you like entropy itself doesn't necessarily we don't entry is a funny term but noise alone doesn't give you any information variations in noise can give you and variations in noise need a reference of none well they don't i don't know if they need that but let's say they it helps have a non-equilibrium steady state to reference against that whole non-equilibrium piece and complex um attractor states and all of that sort of stuff doesn't really feature it's not an attractor state at the top per se it's a it's trans a transition state that some molecules go to or some configurations happen to by chance arrive at so but obviously it's maybe not like that alone there's this kind of inference yes so this reaction would be reversible but imagine if the ball gets to here now it's a 50 50 it could roll either way it's at the transition point that's what enzymes stabilize transition state complexes they reduce activation energy barriers but now if you were down here in p2 it would be a positive delta g so it'd be energetically uphill because this is what they call a thermodynamically irreversible reaction so keq like the coefficient of how complete the reaction is going to proceed is going to be way to the right on this one but if it just looked like a hump in the long run you'd get an even number on both sides it's kind of like ergodicity and so it is there's a lot that can be seen here so dave and then blue yeah the uh advertising meme for that expensive can opener that needs to be pushed that would uh bind to the hump and actually like use i think you said flatten the hump effectively and how does that happen is there i i presumably you're getting a reduction of cognitive dissonance it no longer looks dissonant to imagine oh spending all that money i feel so guilty about buying a forty dollar book the other day how can i justify buying a twelve dollar can opener instead of a two dollar one and the point of sale oh just tap it's lower kinetic friction versus you know pull out your um your shells count them blue and then stephen so one critical thing when you're talking about the whether or not a reaction is reversible or not so you when you have the free energy of the product and the reactant if the free energy is the same it's completely a reversible reaction but the fact that it's an irreversible reaction is driven by the lower free energy of the product of the reaction i think about things like um you know even just the self-assembly so we can talk about self-assembly of life of cells but also self-assembly of molecules right so so molecules will spontaneously like you know salt small spontaneously dissolves in water and that's driven by charge right and hydrogen bonds and you know the polar polarity of water and so when you have this um you know kind of spontaneous reactions that occur that are driven by like molecular charge is that that's again like an information catalyst because the charge is is information is it positive or negative is it a north pole is it a south pole and and so that without that kind of information would these reactions push forward and one way that computers over the last hundred years have revolutionized this discussion is like if you just say well of course information has nothing to do with it we're talking about just chemicals tossing around in space and we're talking about thermodynamics there's no need to bring information into it and then someone says well i'm talking about the chemical components on my hard drive or i'm talking about the chemistry of this processor so it's like there is an information thermodynamic link we are putting energy into this processor or hard drive there is chemistry happening who will link it it's no longer a tenable position to say that there just is no linkage or it wouldn't even be useful at the very least it would be extremely useful if not beyond useful to understand this nexus so it's kind of like that info thermodynamic is already well in progress and we're using devices that are empirical proofs even if the conceptual details are not quite filled in yet blue and then yeah just to respond to that it's like maxwell's demon is essentially can be replaced by a computer right like so a computer can sit on top of the box and decide whether the you know big molecules go this way or the hot molecules go this way and the cold molecules go that way but then you'd have to put the computer in a fridge and you'd have to plug the fridge in right so there's all this like the thought experiment of um you know the information separation or the information filtering taking place um with computers and so that's also like another direct link between the info dynamic and thermodynamic um capacities thanks steven yeah this is really interesting um yeah the this link so what gets really interesting is when it that when there's those endothermic so for instance um actually we used to use this when we used to do reactions in the lab so like if you want to you get a load of ice basically dump a load of salt into it and then your ice goes down to like minus 18 degrees right so now you can start doing reactions at minus 18 degrees in your beaker because you can immerse it in this so your um well what's just happened there right you've just dropped the temperature that doesn't make sense so this is where this entropic you know the ice the whole way the ice melts and thaws and if you know if you drop the temperature very very drop the temperature drop the beat drop the video chat rip steven we'll continue though he'll rejoin um this slide which i just put the image up this is like in the wikipedia article on kinetic versus thermodynamic control one doesn't have to be a chemist to see that we're taking a few very interesting esoteric symmetrical molecules and there's some transition state in the blue if it's very cold it's going to form this product if it's very warm it forms a different product the ratio and the temperature and the solvent there's so many details to explore but it's just an example about like thinking fast and slow type one and type two conomon you know decision making frameworks what's the type one decision where is it really being thought through you know hey chill out cool down slow your horses make the right decision versus buying under pressure time pressure for decisions that's one of the major threats to cognitive security and good decision making is not having enough time or spending too much time on a decision which is implicitly taking away time from another decision because in the lab we can just heat things up cool them down but you know we only get one sequence of action in a broader sense we can't just perform on single decisions because we also need to do that sort of bayesian estimation of how much time should we be spending on this decision blue so just thinking about time i mean time is always the fourth axis right time is always the other other thing but i mean we have these things that we say like time is money right like so so what is the cost of time um and like i think about things like time effort energy and money as all like things that i put in to drive a reaction forward like things that i put in to create a result right time effort energy and money and so is there some kind of like quantifiable interchange between time and money or time and energy or time and effort or i mean it's time always the entropy increasing thing but like so okay entropy increases with time right like i put time into something right so is this um am i putting entropy into something to create a result that seems very backwards um it and it's just cool to think about time as like you know something that you put in to drive a reaction forward i don't know quality time like attention maybe it's like you know if we had a unit an erg or a calorie of attention you know the the the calories the temperature heat of some metal sphere and a cloud you know there's different amounts of specific heat maybe individuals attention will be different it's different to get the attention of peer reviewer one versus peer reviewer two but is there a commonality for attention and does that help make decisions get pushed along the road blue maybe you could introduce this nice set of slides on life-mind continuity sure um so there's a uh maybe maybe the first question first and then we can maybe talk about the life-mind continuity thesis so um this is a quote from the paper it says adaptive active inference requires that a system autonomously engage in active inference maximizing sensory evidence for its own existence this kind of is a different um yeah and they cited this reference and this is from the sims paper but um i just thought about what is evidence for your own existence like when i pulled this quote out and does that require self-awareness like like do i exist i think therefore i am it's very like descartes and um like that's evidence for your existence but like does the little cell it self-aware enough to know that it exists i mean it is like the desire to not be extinguished awareness of your own existence um so i think about you know it's sensory evidence for your model or for validity of your model so that's kind of always how i think about it model evidence is not evidence for my existence it's evidence for the validity of my model so that's kind of how i've always thought about it but it just was interesting this um this phrasing that was used by by sims uh um because i just was curious is there a difference evidence for your existence model evidence and just what do you guys think about that or or am i the only one that was like that's kind of weird there's a lot of pieces here so one piece is that in active inference the preferences are over sensory outcomes not over hidden state estimates instead of maximizing um you know values of hidden state estimates like making it maximally safe out there that situation of bodily safety would be reframed as maximizing evidence for observing our observations in their preferred states so one little flourish that is active inference germain is this focus on the sensory evidence rather than inferred hidden states that is an interesting question about is it is it the model of itself what is this how are we getting sensory evidence ourself and is that the imperative for action and it just reminded me because you said um you know i think therefore i am there this is um uh latin post hoc ergo pro propter hook after this therefore because of this it's like the post hoc fallacy so i think therefore i am and then of course you know cue the 50 fristen papers i am therefore i think every possible inversion like is it that thought begets life or does life be get thought and then if we're only going to find both of them inextricably tied up in actual systems what is the partitioning or the compartmentalizing variously meaningful or true or so that inversion's really cool and and also um speaks to the podcast that was released yesterday on the podcast episode where um it's clipped from i think live stream five six point no 5.2 there's a lot of discussion about change your mind change your life and then the inversion of that is change your life change your mind right i i think therefore i am i am therefore i think it's just that's all this action perception loop of active inference yeah here's the first in paper i am therefore i think he he did go there he's been there this is an interesting um an interesting uh paper it it even begins um the first sentences how does schrodinger's question which is what is life touch on philosophical propositions such as descartes famous proposition cognito ergosum i think therefore i am and then a little not exactly a formal connection but um this weekend is kognitio 2021 kind of another illusion there but yeah what does it mean for sensory evidence of our own is that our um our proprioception and our our uh thermo perception and taste or is this referring to a higher level of mental activity like as we saw in the san fed smith paper on mental action the observables of one level so like the outcomes of the higher order mental model can be plugged into the priors of a lower model or the hidden state estimates of a lower model so like is observing our attention maximizing evidence for a model which we pay attention to different things uh where does where does that play in with life mind continuity what is that question oh gotta hit unmute i got it so um the the distinction that sims made in the paper between adaptive active inference and mere active inference so my question is whether this distinction follows the kind of evolutionary late comer path or if or if it's in line with the life-mind continuity thesis and so we can kind of unpack all of those in the next couple of slides okay so the um life-mind continuity thesis so i i i pulled from this paper by kirkoff and froze for what you see where they asked the question are mental phenomena restricted to living systems and there's several viewpoints here the the first one is um obviously you can answer no and there's a couple of different um ways and and so the cognitivist viewpoint uh is that some cognitive systems can be living systems but mines are really associated with computational processes that have semantic contentful properties and minds can be realized independently from life given the right kind of artificial support system and so this this allows um mind to exist artificially essentially uh and then there's the overly generous non-cognitivist free energy principle um thought that that all systems can maintain their variables within a limited range of values all systems that do this can be understood as having some kind of mentality or proto mentality given that the fep casts any system that maintains structural integrity in a fluctuating environment um as engaged in predicting its own future states and this is kind of the pan psychic take on you know that life can exist in or that not that life but the mind can exist in things like molecules that are undergoing reactions so there's i mean in everything there's a mind thanks for this really helpful breakdown huge topic you know a lot of implications is it okay just to throw out a cell phone that we're done using or would that be you know a a rights issue so are mental phenomena one-to-one restricted necessary sufficient how is cognition related to life as we thermodynamically know it and some some strong shots from the no side which is saying um which first off is always a somewhat easier position to defend because you can just say look it's a venn diagram overlap and we don't know so who's to say that it's restricted because maybe there's some overlap maybe there's a but not b maybe there's b but not a so first it's a general claim that one can deny a restriction by just saying we're not sure or it might not always be that way and then this is just really it's it's clear and funny how you've written it overly generous like relative to the expectations of who maybe someone's very stingy with their fep interpretation maybe someone else especially the authors how they wrote it so you're confident ferguson that's how that was their their their um terms nice so maybe there's an fep whale out there for whom this is still not generous enough but um i i i if it was cool i didn't mean to to edit it um but there's uh the integrated information theory people flavor of pan psychism would be like because all systems are on this continuum of integrating information pan psychism another approach is sort of this cybernetic pan psychism which is like because all systems are on the continuum maintaining their variables within a kind of like good regulator theorem in other words not because there's an information continuum but because there's a cybernetic control continuum dot dot so there's an information centered inference pan inference equals pan psychism and here's like a pan control or pan resistance to dissipation equals pan psychism claim so that's pretty cool how that that action and inference thread come back um in different guises welcome back steven your surroundings are less but i hope that it will work yep we're on 11 we just heard some of the no uh voices on the question or mental phenomena restricted to living systems and then who is on the other side of the table well and then there's a couple of different um views that the authors put forward in the paper as well on the yes side so there is the non-cognitivist fep so that's the the second view from the no but plus um evolutionary late come review so this is mind but not life requires sophisticated generated generative neural machinery that is not present in simple forms of life such as single-celled organisms and that was my question about adaptive active inference versus mirror active inference is that like putting mind above a certain is that at a certain evolutionary point such as like higher level vertebrates or whatever only perform adaptive active inference if they're evolutionarily at a certain point or in a certain as sims kind of implied in a certain amount of collectivity like so so uh in his paper he said that you know whole organisms can be can do adaptive active inference but not like cells within an organism and so that was kind of it's along the lines while it doesn't necessarily um link directly to sophisticated neural machinery it seemed like sophisticated collective autonomy or like like what what was the the sophistication and and where was this line and i kind of pushed matt on that when he was here but i just wonder is that is that linked somehow to this way um and then i don't know stephen did you want to say something or and then i can talk about the life minecon continuity views yes yeah um i think actually it can just kind of continues what you're saying continues a little bit with the graph that we had earlier there in the sense i was thinking when when um something goes over that that graph and goes down the other side um with active inference the like with normal reactions right you do it anytime with the same reactants it's the same basic curve every time right because it's it's a standard thing but with life it gets a chance to know what the final outcomes have been and start to maybe change the energy of the hump it can actually change particularly maybe as well if there's non-equilibrium steady states in there

there's an actual ability to infer backwards and i suppose that that inability to infer backwards well in real time which in real time could be chemical time that that could kind of happen through enzymes or maybe through the mediation enzymes and whether that's cognition or whether that's could be done through some real-time um you know chemical inference process um and then of course you've got that happening over longer times um and this then becomes when do you start to move from a variational free energy into an expected free energy you know um but that ability to go backwards or forwards and start to sculpt the landscape of intermediaries um yeah i wonder i think that kind of ties into what you're saying i'm gonna do a little life-mind venn diagram what kind of a universe or world or simulation or whatever do life and minds look like partial overlap like in this world let's make this a little different color you know in this world there's things that are having mind but not life and vice versa is a strict refutation of the restriction claim is it restricted to living systems this would be no but then let's just say it is restricted to living systems well it could be like that there are still living systems with no mind or could they be the same so how would we know what would be the observation or the experiments or the measurement to make especially keeping in mind or keeping in life that mind is a little bit of a potentially unobservable phenomena so how do we integrate sort of meat and potatoes with minds in a way that's going to lead to testable predictions and optimally informative experiments blue well and so that is the the life-mind continuity visa thesis is that um there's it's 100 so the concept of a mind is grounded in the concept of life in terms of autocoesis adaptivity but the problem that i have with that because the whole reason i brought this up is like back to schrodinger's questions what is life right and so um it's not binary necessary necessarily like there's some continuum between life and not life there's a lot of gray area you know obligate parasites there's viruses there's a lot of things that fall into that like fuzzy area and then also like going into the origins of life theory somehow some self-replication and autophysis happened in non-life that like drove the transition to life at some point in our evolutionary history and where's the mind there and that's one um other reason that pan psychists implicitly or explicitly promote their viewpoints is it there's no need for like a biogenesis for a mento genesis for the emergence of mind from non-mind you just say nope my belief is it's all continuums of mind so we're just like adding grains of sand to a pile there's a debate about when it's a heap versus when it's not a heap but there's no qualitative challenge with explaining the emergence of complex minds as described in gerda escherbach whereas other people who are not pancyst implicitly it's like saying you do need have an account for how minds if you think they exist arise from non-minds or you can take the anti-psychic position and just say there's never emergence of mind so not in the little not in the big and so there's a few ways to be consistent you know pan psychism and then elaboration of mind mental denialism which is its own answer or some sort of qualified intermediate where you think that it emerges at a developmental stage or some evolutionary development or an ecological context or some informational requirement you have to be this many bits to have awareness so there's a lot of views in the space but this is just total fundamentals what systems are living which ones have mind which ones deserve to be treated using respect steven this also brings back the multi-scale maybe not necessarily i don't know if physical scale but temporal scale you know if something is actively variationally enabling itself to um maintain its steady state um but it's purely doing it only at the level of maintaining a chemical gradient and that's that's the whole game i don't know if that's a bit it depends when it's not exactly clear where mere active influence active inference sits right so there is a slight ontological question there but the if something's maintaining itself but it's just doing one thing and it can do that maybe through chemical gradients or whatever but then you've got this temporal depth okay doing that because it's inferring that by doing that it will allow it to do something else at what point do these kind of chain of events start to become more life like and does then then need to be is that in itself life and does there almost need to be a lifelike entity that is managing that and then that becomes mind and then this is this is that question that i suppose fungi raise a bit as well because they they mediate messaging part message passing and a lot of interesting stuff now are they kind of a mind or not uh i don't know um we could definitely go to talking a little more about um mirror and adaptive active inference but it makes me think about you mentioned a chemical gradient well what are neurons and glia doing if not shuttling around chemical gradients and so it's like okay is mind in the sodium ion okay you know yes or no garden of forking paths is it in the system of the lipid membrane and the transport protein and the gradients it's like nope that still seems like a fairly no mind needed physical system okay well now we're gonna put some other bits of protein around it and now we're gonna converge towards biosemiotics with signaling at the membrane but it's still a purely physical system and so it's you start building up and is there anything left to explain is that reductionism to just sort of make everything literally reduced to the explanation of the chemicals and it reminds me of a of a sapolsky article in 2004 where he says if if free will is lurking in the in the uh interstices of the brain then the crawl spaces are getting smaller and smaller because as we get tools that help us look closer and closer we're finding there's less and less wiggle room degrees of freedom tilly lally or whatever you called it less and less of that with deeper and deeper investigation and dissection but are we just tearing something apart not understanding how it's actually composed blue so that just reminds me of back going back to the live stream that um we had with mike levin if we are just at any given point our model essentially like dictates our action right so in under active inference we have the degenerative model the recognition model and constantly these are updated based on what's happening our world can we in fact or do we have a choice when presented with any given circumstances in life in our environment do we have a choice whether or not to steal a candy bar from its tour or like like random things that you think about um this also goes to change your life and change your mind so it's only through like practicing piano do we learn to play piano but do i have any choice but to sit down and practice the piano given my existing model or and so is there like at what point where does free will come in like right like does it play in and um mike's answer was awesome i think he said like no there's no free will but then he said okay there's some minuscule amount of free will that's non-zero but like the smallest possible non-zero number that you could have and so incrementally then you're able to effect changes on your model which then affect the decisions that you make given your model just two quick things then steven so with a chaotic system with a big leopard of exponent so small changes have non-linear effects all you need is a tiny tiny bit so let's not um confuse the fact that we have small but potentially even infinitesimal or fractional free will whatever that means in the moment with our inability to have serious consequences through deep time because it is a chaotic massively interacting system and the second point is it's um you mentioned like stealing something that 2004 paper with zeki good enough and sapolsky is about the frontal cortex in the criminal justice system and this paper is really fascinating because it's like there's two modes to just i believe centering on the american legal system but replicated many other cases either somebody can be like um a rational sane actor or they plead a defense which basically amounts to being entirely incapacitated so like either you were fully you should have known that um the bank robbery would have gone this way you should have really waited it out all the ways it was a situation of which there are multiple where you cannot be held to be responsible for the action but we still need to you know it's not to get out of jail free card but it's it is what it is and so they're arguing for a more nuanced perspective that respects all the different time scales of changes that influence our decision making so it's a really and and you know this is 20 almost 20 years ago so a lot has changed in neuroscience and in law and society but this is a little bit of an older peep now i think this is quite an interesting point that you're making here i i i on that slide if you go back to that slide that has the mind on it i'm just trying something here if a little idea i i've just got it if okay you say so if i was to put life you see there's life there put life over mind okay so i'll do actually just do that so it's like this okay so say this is i don't know this is an idea so say you've got life as in the organism's body and then you've got life in terms of the the the the cognitive neuronal they're like they're differentiated into two types of organism in the organism and the two organisms talk to each other and the boundary between the two talking to each other is mind as an emergent something that's coming up as an idea interactionist and relational and interfacial that maybe we shouldn't be looking for the reduced essence of atomic modular life but we should um relentlessly pursue it in the interfaces dave yeah the um if mind is entirely a matter of memes and memes need material processes to live in and material processes need means to keep themselves from falling apart then things become a lot simpler uh but i actually want to make a comment on the insanity plea i think it might have been dershowitz who went back and looked at the statistics of successful and unsuccessful insanity pleas and he found that when people when somebody is so completely berserk that you can't even begin to empathize with him juries say he's just crazy kill it but the places that people the juries by the the insanity plea is the juror saying to himself although he never quite comes out and says this under the circumstances i could see killing the guy that's all right whatever it takes to get him off uh you call that insanity okay we'll get him off with insanity so it's exactly the opposite of what you'd think you know the supposed elizabethan argument that well the guy didn't know the difference between right and wrong so you can't hold him responsible american juries don't think that way big questions about what the legal system is for and what it does flu and then there's a question in the chat so just on to like the legal system and i i wasn't like meaning to start spawn this entire conversation with the stealing candy bar but but you know so much of our creation of the model of the world occurs when we're young and in early development and and i think that um you know at least in the states the the amount of un unacknowledged childhood trauma that happens um for poverty and for drug abuse reason i mean there's a infinite number of reasons that people undergo some kind of abuse as a child all of that like really contributes to how well you're going to function in society in later in life and i don't think like while there there are people that plead insanity and there are some cases of of severe abuse and whatever where the people get off but but so much so many of the people that are in our prison systems and go through juvenile systems have these horrible early experiences that really are happening when their model of the world is being solidified and so for me it really um it explains a lot of the choices that are made um by people in those circumstances and it's helpful to think about it in terms of a trajectory we're not just um maybe taking a snapshot moment and just doing a snap judgment but we want a trajectory of healing and of um health stephen and then dave and then we'll ask the question yeah this idea because we talk about people being out of their mind you know and in some ways like you say these if someone has an experience or something some biological condition the idea is it's like i suppose the biological architecture means that the person's ability to mindfully choose isn't there in some ways it's almost like the yeah either the the somatic priors are so skewed or the uh the mental architecture is so distorted a coherent mind isn't available thanks steve i met a lawyer who worked with the um criminal justice system in chicago on um child abuse cases and she's had a huge number of these cases of serious abuse like a three-month-old being put out on a fire escape in the middle of winter to toughen him up and she know instead of just saying oh this is so horrible you got to put the mother in prison they interviewed her says what do you do with babies and she had no idea she had never seen even minimally competent child rear from anybody and they found you explain to these folks you've given you know the supplemental education to understand why middle class people treat babies a certain way they think that's great oh that makes sense i just didn't know you could do that i thought you had to beat them they wouldn't stop crying if you don't beat them and say you know you put them on a long time probation that is not punitive at all just so you can keep track of them and keep the education going and keep them away from the crazies and just it was working extraordinarily well thanks so here's a question from the chat i'll read the question and then i'll read one response in the chat already and then um anyone can jump in and we have some slides that speak to this so joseph wrote could the extent of how conscious something is be measured on the basis of how far ahead in time an organism cell creature can plan or minimize expected free energy and then i know that some people will probably have things to say on that but tucker in the chat wrote under this idea then might the entropy chaoticness of the environment influence the degree of consciousness of a system a more dynamic system or environment would require a more adaptive agent in order to maintain negantropic coherence through time the more entropy the need for a better model super interesting questions and responses so i guess just uh glancing off of that uh legal and choice discussion into more broader questions it's so fascinating how especially if we're going to be locating function in the interface between the agent and the environment maybe depending on whether you're in um an unmarked room or a very interactable space consciousness is going to be different are you more or less aware and then the other piece that comes to mind and then stephen and anyone else is um from livestream 25 computational boundary of a self and um there's the past and then there's the future below and so the past cone is kind of like the the depth and the breadth of what has influenced the system and then the uh the future cone is about the scope of planning so it's a really interesting question about temporal horizon but is it enough to just plan deeply like if you just plan sparsely but deeply over a hundred years versus a very rich plan very thick plan for 10 minutes which one of those is there gonna be a single currency that describes or compares those two how do we think about planning in heterogeneous systems have different affordances or have different complexities of their internal generative model so there's a lot of great questions in this area stephen then anyone else yeah these are really good questions um i suppose if we're looking at measures i mean there's also this interesting question of not necessarily what you're able to do but how far you're able to get with what you're not able to do like the like when someone's really working hard cognitively it's like like i mean if i want to predict what the inside of a jar is going to be like i can just boil the liquid and it becomes sterile and i know it'll be like that for a long time um the question becomes when you've got these intractable kind of multifaceted things a bit like what's going to happen to a a collection of family members when they all meet together for thanksgiving right um this that that that that temple depth is is definitely a good point and i suppose there's that question of um how to penetrate when it's hard to penetrate as much as how much are they able to plan but i'm sure the two things are linked and you could do both blue and then i'll have a thought uh give your thought first daniel and because i'm gonna go on a tangent okay i wanted to connect it to the paper that we're allegedly discussing and this user resource and mirror and adaptive active inference so on the uni-directional the left side you have like a more adaptive agent it's here a spider and then you have this less adaptive but doing mere active inference like a web okay so it's like a person with a thermometer or a spider with their web versus this reciprocal case where on both sides of the table there's a user we've heard before about how the time horizon of planning relates to the um the depth or the quality of the conscious experience for example the 2018 alias interview with myself and martin and carl fristen in the final question he even talks about the temporal scope and the existence of counterfactuals as being relevant for generating consciousness so then it made me wonder what if by going reciprocal multi-integration you actually shorten the time scale of planning like if i'm gonna um set out a buffet for myself it might take hours of planning and so there's a lot of um mere active inference happening on the stovetop but it's a very complex process one user and takes hours of planning but then to cook with a friends might be just a minute of planning we're going to agree to cook together and so actually the temporal explicit depth of planning in an improv improvisational or interactionist context the depth drops potentially because you might not need to explicitly prepare for so many counterfactuals and it's not like you're just moving pieces on the chess board you just need to take that first step listen to what they're gonna say then think about what your response is so then that almost challenges simple time horizon or counterfactual basis of cognition or phenomenology because um and it's a such a helpful distinction made by matt that i know we're going to use in other papers and research that these are really different kinds of systems this left system is stigmatic and it's still a collective system with emergent properties but then this kind of double stigma g interactionism brings us into a new territory that's very important to delineate yeah this and this takes us into that realm of sense making different types of sense making because it's action yeah being able to act and interact in appropriate ways and in many ways the more appropriate ways can be stimulated by things in the environment i mean the fact that we have a very expressive face in a way puts our face out there as something we can inscribe and use in that process right we just look at each other's faces now for all the psychology in the world no one quite know you know it's it's it's all it's all it's all an interpretation but the the thing that i think this really highlights is the game in town is action policy it doesn't the game isn't for us to walk around as humans with big cognitive capacities on the top of our skull or in our skull the game is that we can interact and sense make um the ability to like you say to plan on our own is um is a nicety but actually it's been a bit of a we could say a wrong turn for the last 200 years because it's all been about thinking as being the game and feelings being this kind of artifact that we've we try and minimize as some sort of side product that's somehow you know a weakness of the human mind or something and and it's so far from the truth because it's pretty much what it was like before we took that wrong turn it's actually about how to act and feel in the world so this is a good point so i'm going to go off on my tangent now but i really just want to address the first question in the chat about how consciousness relates to our ability to plan in in how far out that we plan and and so i'm reminded of and maybe daniel you can help me remember if it was deeply felt affect or sophisticated but in in one of those papers we talked actually anxiety as trying to plan too far out in the future right so if you're trying to plan too many time steps ahead then that that is like the cause of a neuroatypical situation so i'm not sure i will and then i think that they said that maybe depression was staying too many steps behind or thinking about too many uh steps in the past so thinking about um and it was so my takeaway from that was staying in the present moment and so i don't know that the that um just briefly to respond to steven's comment i don't know that um having this big cognitive giant capacity giant cognitive capacity is for interacting as much as it is for adapting um because humans like cockroaches adapt to you know infinite circumstances right we live on all or are present on all continents in the world and and i mean even antarctica there's some research scientists there right now i'm sure digging holes in the glaciers to find out how much our ocean is going to rise i hope that somebody's there doing that anyway i'm not sure who's funding that research but but hopefully somebody's doing that i don't know i think it's more adaptive the cognitive capacity than anything and to so good good throwback there with the uh sophisticated and deep active inference and the anxiety the parametric investigations of anxiety that we've been on before and one day when we have better keyword indexing if anyone is passionate about they should collaborate with us because how cool would it be to have deeply richly indexed keywords for these papers and artifacts and discussions and have them translated into multiple human languages and computer languages so and graphics so that we could be really clear so i mean there's a lot to say here and the only other throwback that i was going to add in was to way long ago in number 29 we talked about the scaling relationship of the ants brains with a larger colony having a proportionally smaller ant brain whereas larger mammalian societies have larger mammalian brains because mammalian sociality has to do with like memory of repeated interactions and theory of mind and the facial recognition all these things that we're talking about here whereas potentially colony organismality means that the increased scope capacity of the colony can occur over evolutionary times through a diminution of the scope or of the cognitive processing on board each ant so joseph's question it's sort of it's just such a great question so thanks again for asking it we can sort of look at it at the i mean it's even asked across scales but what if like the cell plans less because it's it's part of a high trust system so now the the beta cell in the in the pancreas doesn't have to have a deep temporal model its temporal model just has to be like one step but it's playing a role in a system that now has a circadian regulation of sleep or a blood sugar and so where do we ascribe consciousness or awareness or cognitive possibility across different systems which again brings us back to um from 25 these nested cones of cognition and their different shapes and their attributes blue um i just wanted to shout out to tucker as i was like i don't know drawing this big thing uh with the last question so tucker actually made the link between consciousness and adaptivity and i kind of just stole it i stole all the thunder um but but really uh and are those two things linked right so so planning very far in the in the future versus the having the ability to respond to whatever comes your way in one time step those are really different like one seems very rigid and the other seems very fluid and and tucker actually um kind of used consciousness and adaptivity interchangeably which is an connection and thought um i don't know maybe resilience and that adaptivity is is kind of what i would use but but it's interesting to think about yeah and also i like that point you're making and suppose it ties with that adaptivity about the pancreas um you know you if you've got the pancreas gives you the affordance of modulating stuff in the body as do these other uh organs but like you say these cells if i have a crude or multiple ways of mindfully or livingly i don't know if that's a word adapting that you can do something but you got no pancreas or you've got no cells that give you an affordance to actually change it you can have all the computational cognitive modeling whatever it is you can't do anything it's literally a conceptual framework without any way to do that sort of control so you know the more important thing is to have something which can follow through at these across the scales change how they are maybe cybernetically integrated than it is to have the like i say it's not all about cognition it's about action policies and i think that's really interesting so earlier we had just those three right just cognition agency and autonomy and then that was um being inspired by just some of the quotes from the paper where those terms were used but we're thinking about that three then we've talked about like adapting and being flexible and learning over uh developmental as well as evolutionary time skills we can think about the past and the future like the time cone illustration to what extent does memory matter i mean can you have a memoryless awareness or is awareness requiring memory or can you have awareness without planning some meditation practices might lead one to think so or is planning vital what about the capacity for interaction if we don't have interactions highlighted you know paging dean are we falling victim into a modular reductionist nightmare or how do we hold interactionism at every step what about kind of complex systems properties like resilience or anti-fragility or hysteresis phase transitions those are descriptors of various systems and are those just coming along for free with these do they grant the ability for these are they correlated dimensions are they uncorrelated million dollar question maybe we need to scale that for inflation you know the 10 bitcoin question where does consciousness come into play and is that something we can ever get a handle on or is consciousness always going to be at the fringes of our explanations and some of these more accessible or opaque features will be amenable to quantitative descriptions whereas this um might be always in a different category so blue well i think that they're quantifying consciousness already but um memory is really what i wanted to talk about here the key word that's on this list and so i mean i don't know i think about memory right like there's the rom and there's the ram is memory or to what degree is memory required for a generative model i i just wonder like i mean all of the experiences contribute to the formation of the model and what is contained in the model is effectively like our action course i mean what's in the generative model does anyone know um i think about it like like what is um the actual i think about it like an action course or like the response i don't know i'd love to hear what other people think it's in the generative model what's inside of that um and also um whether or not memory is required i think this also you can bring in there with like affordances coming they say you know like you say interaction capacity with affordances uh i suppose you could say memory is an interaction capacity with the affordances to whole priors you know they um so there's an interest in um there's an interesting sort of dynamic here between these processes and the affordances normally affordances are well for practical purposes are normally thought of or used in terms of something a bit more concrete and at the human scale but when it kind of is multi-scale as well isn't it you know if we're going to talk about mental action and thought as mental planning then absolutely there are mental affordances in the model as formalized by sandved smith and others and then to blue's memory question about where it's related to the generative model so i thought of a few different options not exhaustive so the generative model could have an implicit onboard memory like that d that could just be a summary variable that summarizes memory so you can just say the past moments temperature here's my prior on the past moment's temperature the past sequence of temperatures could be explicitly represented as like kind of like a list or a vector so there could be on-board memory like within the internal states the internal generative model of the active inference agent could either have just sort of a snapshot summary implicit memory which is good enough in many cases so we might just model it as if anyway or there could be this explicit just like a policy is a sequence of actions you could have a multi-time step prior over previous states aka memory but then the other option is like this extended or stigmatic memory with a recognition model like do i remember what i looked like at that one family dinner when i was three no would i recognize the picture yes would i know where to look for the picture yes physically what about asking somebody so we're thinking okay affordances social affordances spatiotemporal location of extended cognitive process recognition but do i remember that or am i merely remembering or preferring certain affordances that bring me to the edge of that abyss and then the recognition model takes over once the picture has been physically presented that's the stigma g just using matt's distinction that's there's you in the photo library that's the unidirectional multi-scale inference integration and then there's the conversation where we actually spur each other's memory uh and that's where we get that sort of next level of reciprocal integration to kind of bring it back to that idea blue really nice with the extended sigmoidy recognition model um and i think that like there's some like ram some kind of ram is required for any generative model like like i think about the generative model a lot like the software that we're running and in order to have software you have to have some ram um but but is the long-term memory required also like so or is the long-term memory just what goes into creating the generative model and then all of that can be forgotten and and what i'm thinking about that i'm thinking about people that have lost their memory for whatever reason um like so so cases of amnesia like the person can't remember anything like i think about um the old movie memento and like you know every day i can't remember anything i can't remember anything but like i still know how to brush my teeth or i still feel like when i wake up in the morning i need to brush my teeth or i need to shower or i remember what food is right like so i don't need to like you know try this or that to see if it's edible again so there's some things that are maybe stored in the generative model and maybe some things are irrelevant thanks blue steven yeah this this is um ties into the action policies of remembering like if all memories is an act of imagination is an act of recapitulation what needs to be present as you're saying is is a way to evoke affordances and you could say that the affordances of remapping uh sensory states you know maybe where you look with your eyes and maybe when someone's amnesic it's not that they haven't got the priors or the knowledge they haven't got the action capacity to recapitulate from whatever priors were there i mean this is a bit speculative but uh anyway it reminds me of the uh neuroscience on taxi drivers and spatial reasoning and general relationships between spatial memory and practice there and cognitive capacity and preventing cognitive decline and memory palaces like memory palace is kind of this weird quirk i mean how weird you can imagine a place with file cabinets and walk around it and have stories about that place and remember you know pi to 5 000 digits or remember the iliad how weird oh wait spatial cognition narratives of action you can build that physical palace and have the photos there you can update your generative model to include a spatial context where those cues exist and so it's just a very interactionist way of thinking about architecture including mental architectures like you just need to have the exit sign and then when somebody wants to leave they're going to recognize it and walk towards it and then the next step will be there and so the architect puts the next step there safely and accordingly to function but when we're the interior mental designer or architect we can make spaces that also prompt us and so it's not like we're storing the memory itself in that location were remembering a very compressed version of the cues that interact with us to then generate um upon demand yeah that's a good example the memory palace because is is the that when something's laid down in such a palace if you go to visit the palace and you get this whatever sensory cues that come up it's um it's solid enough that you will come back up with us with a particular answer you know what i mean you'll remember x y and z from that so it appears like you've remembered the thing so to speak but actually it's just that that combination of bayesian inferences will lead you to act to recapitulate it in the way now obviously the average person can't do it as well as these trained experts but and they they can actually do it in a very sophisticated way because the in terms of pure floating ideas is that whole they say six plus or minus one is all you can keep in your brain at any one time in terms of a just a concept that's floating and that's uh but then that obviously that that isn't what's the case when you can start to use memory palaces so it's a very different thing going on it reminds me of um one class as we go to our closing notes a class where just each week the homework and class was to memorize a poem and then recite it that's kind of how education has often proceeded and it's something that um is just fundamental for oral cultures and has been a skill that is less emphasized in literate cultures there's that dense cross-linking of like rhyme and narrative and tonality and rhythm so that you really don't need to keep 50 things or 80 lines in your working memory in your ram with a computationalist metaphor what is being remembered or recalled or prompted and that investigation well if it's not being just stored as a text file what is it and then it's just so cool how it dovetails very nicely with deontic cues and ostensive cues in active inference and conversational theory and a lot of other areas blue and then we'll have last comments here so just what you guys we're both talking about um reminded me of cal newport's book deep work if you haven't read it i highly recommend it it's awesome but in that book he talks about um creating a memory palace and learning how to memorize a deck of cards as an old skill that people just don't have anymore so you construct the memory palace you learn to have a set of cards memorize instead of like you so you can literally memorize 52 things in your memory palace anytime so you can memorize the order of a deck of cards just like he says to build this as a skill because it's enhancing of memory and i wonder if this isn't going to be something you know as we're becoming more and more illiterate um which is it's an interesting i think we've reached like the crescendo of literacy at least in the in the us um and have we reached this crescendo because of technologies like this like i don't need to read that paper i'm just gonna listen to this group of people talk about it right and get the key points out of the paper and their summary and their discussion and then i'll have some interesting things to say about that paper um and so is it is it art kind of technology that's paving the way for illiteracy again and will memory become increasingly important is great to think about and i'll just give my final thought here um thanks for the interesting discussion today and to matt also for the for the provocative paper and i'm looking forward to reading and discussing more of his work thanks blue steven or dave any last comments yeah i do quickly thank you again a really good conversation and uh yeah i think there's definitely something here around maybe the idea of adjacent possibles as well so was there something about the memory palace that one thing is then an adjacency to the next thing when you have to hold something on its own it's going to have to be supported by its own infrastructure i suppose and that's quite an overhead so yeah because there's also this thing of how long you can hold something in your memory without having a way to ground it it just sort of fizzles maybe for some people quicker than others but uh so yeah all very good i'd have to go back and think over a number of points came up here that we could expand on in the future thank you stephen dave any last comments nope just gotta followed up on these references thanks all thank you well to matt sims for writing this great paper and joining our dot one and watching the dot zero for um dave and blue and stephen for joining and for for helping in different ways on the dot zero brings us to the end of and next week's at this same time we're going to be in 31 non-equilibrium thermodynamics and the free energy principle in biology by patricia palacios and matteo colombo so at least one author will join at least one week stay in touch let us know if you want to join for 31 or beyond and see you all later